

DRM Model Selection: SNR and Distributional Robustness

Nested linear-regression simulation

Nested Linear model

$$M_1 : Y \sim X_1, \quad M_2 : Y \sim X_1 + X_2 + X_3.$$

$$Y = 1 + \beta_1 X_1 + \delta X_2 + \delta X_3 + \varepsilon.$$

The simulation is divided into two cases because they answer different model-selection questions.

Case 1: the simpler model M1 is true

$$\delta = 0.$$

Both models contain the true conditional mean, but the larger model contains two useless predictors. This case measures whether a selection method can prefer the simpler model when predictive distributions are very similar.

Case 2: the larger model M2 is true

$$\delta > 0.$$

The simpler model omits relevant predictors. This case measures whether a selection method detects the additional signal contained in the larger model.

Sensitivity Check

Signal-to-Noise Ratio

The total signal-to-noise ratio is

$$\text{SNR}_{\text{total}} = \frac{\text{Var}(\beta_1 X_1 + \delta X_2 + \delta X_3)}{\text{Var}(\varepsilon)}.$$

For nested model selection, the more relevant quantity in Case 2 is the discriminating SNR:

$$\text{SNR}_{\text{disc}} = \frac{\text{Var}(\delta X_2 + \delta X_3)}{\text{Var}(\varepsilon)} = 2\delta^2.$$

Changing a **coefficient** (effect size), **predictor variance**, or **error variance** can therefore be described through SNR.

Distribution shape

Changing the distribution shape of a **predictor** or **error** term while holding its mean and variance fixed, to test the robustness of our strategy.

We generate data from 4 distribution shape with fixed mean 0 and variance 1.

distribution	mean	variance	skewness
normal	0.0019	1.0013	0.0005
heavy_tail_t5	0.0020	0.9961	-0.0108
right_skewed	-0.0031	0.9827	1.9484
contaminated_normal	0.0050	1.0159	0.0320

Research Design

We use five DRM metrics with basis function:

$q(y) = (y, y^2).$

And we also introduce two benchmark: AIC & RMSE.

For each scenario, one **test set** is generated before the **500** times Monte Carlo loop and remains **fixed**, Only the training set changes across iterations.

Representative nested scenarios

$$X_1, X_2, X_3 \overset{\text{iid}}{\sim} N(0, 1),$$
$$\varepsilon \sim N(0, 1).$$

1. simpler model is true

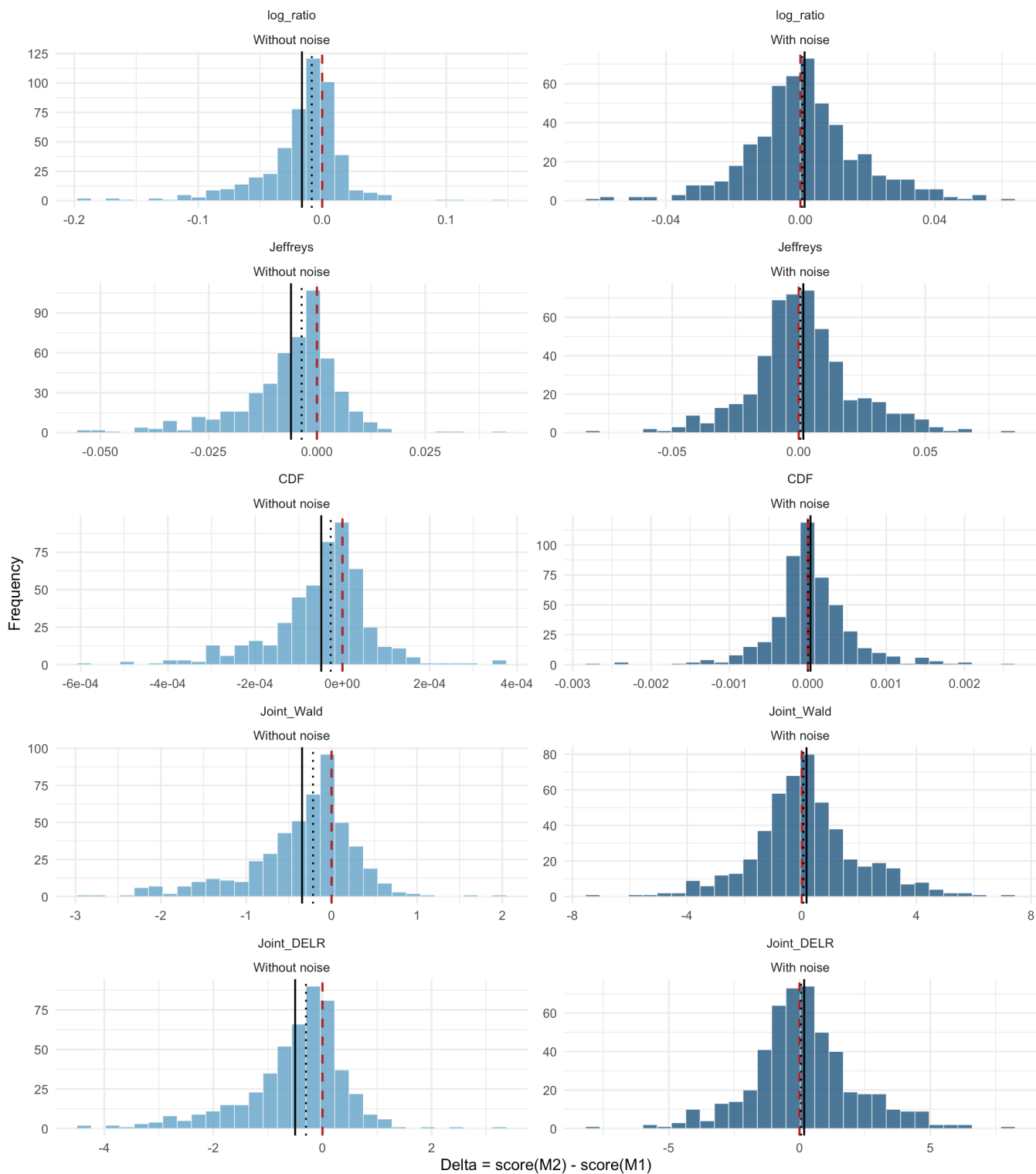
$$Y = 1 + X_1 + \varepsilon.$$

Representative 1: correct-selection accuracy

Prediction output	log_ratio	Jeffreys	CDF	Joint_Wald	Joint_DELR	AIC	RMSE
Without noise	0.286	0.286	0.328	0.288	0.286	-	-
With noise	0.522	0.516	0.514	0.514	0.516	-	-
Benchmark	-	-	-	-	-	0.844	0.622

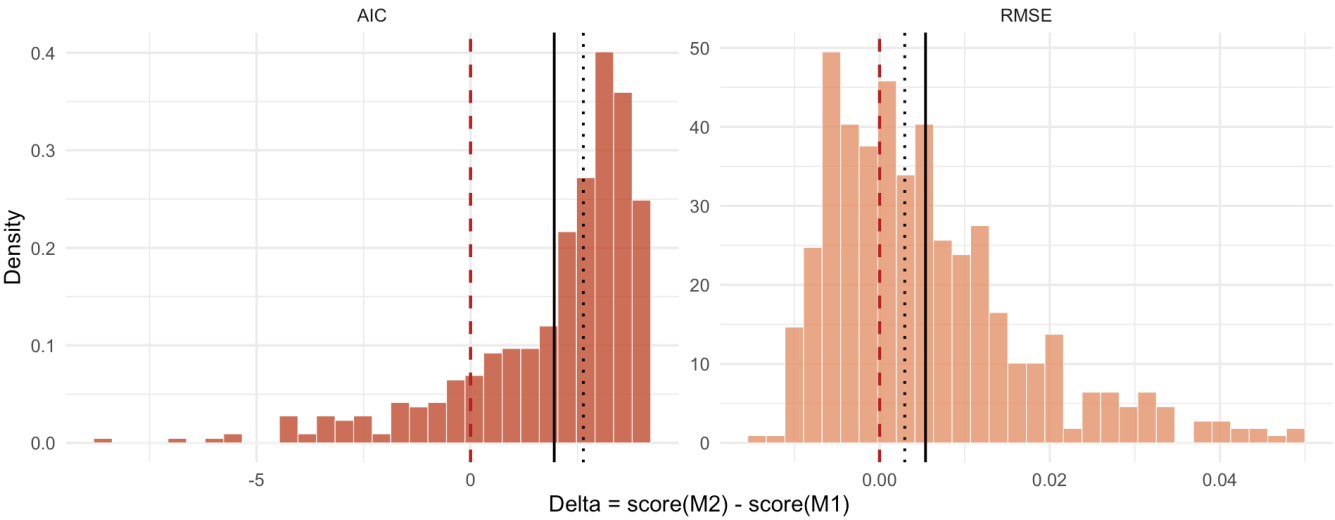
Simpler model true: DRM Delta distributions

Delta < 0 favours M2; Delta > 0 favours M1. Red dashed = 0; solid black = mean; dotted black = median.



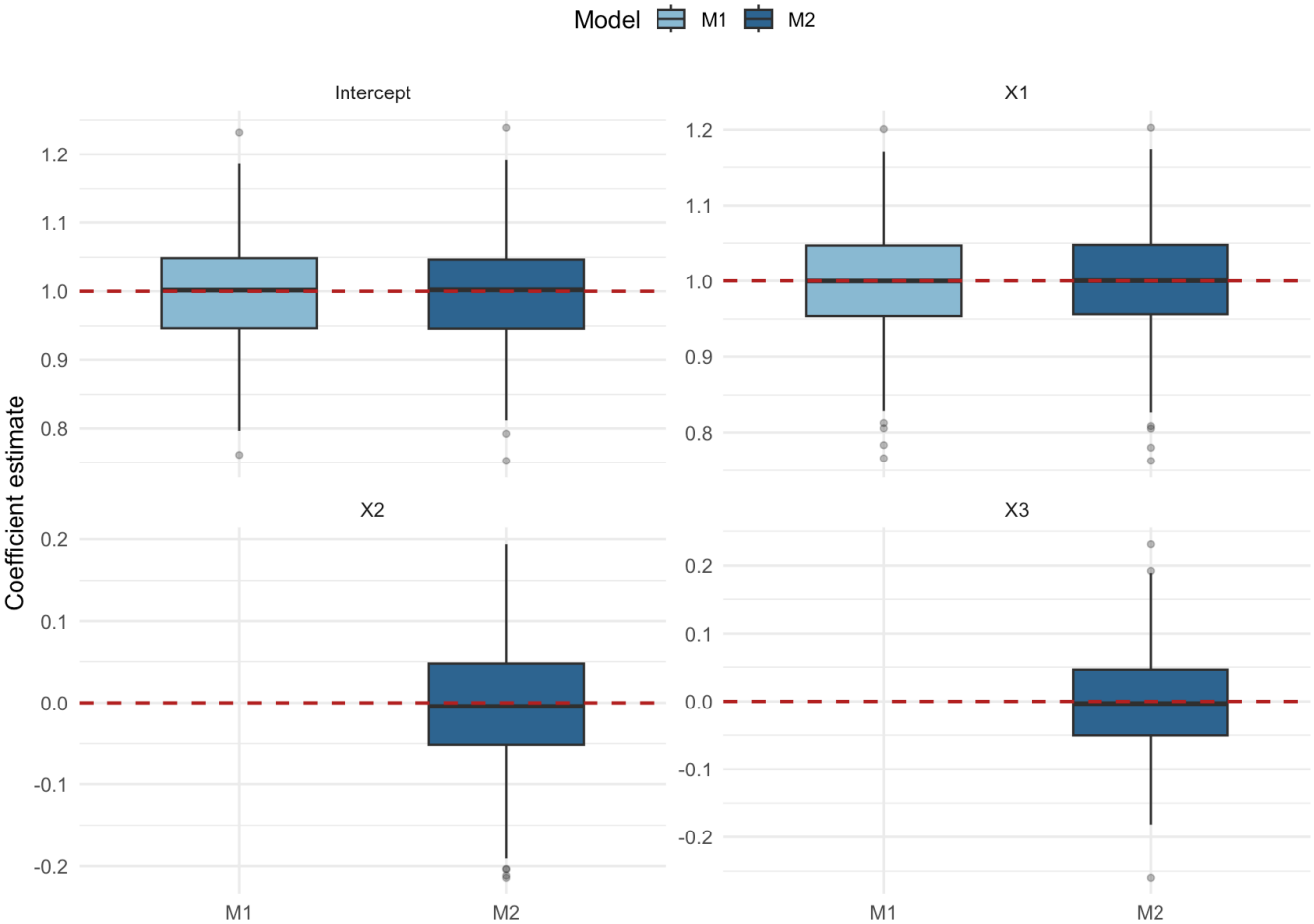
Simpler model true: benchmark Delta distributions

Delta < 0 favours M2; Delta > 0 favours M1. Red dashed = 0; solid black = mean; dotted black = median.



Simpler model true: coefficient estimates

Red dashed line = true coefficient



2. larger model is true

$$Y = 1 + X_1 + X_2 + X_3 + \varepsilon.$$

Representative 2: correct-selection accuracy

Prediction output	log_ratio	Jeffreys	CDF	Joint_Wald	Joint_DELR	AIC	RMSE
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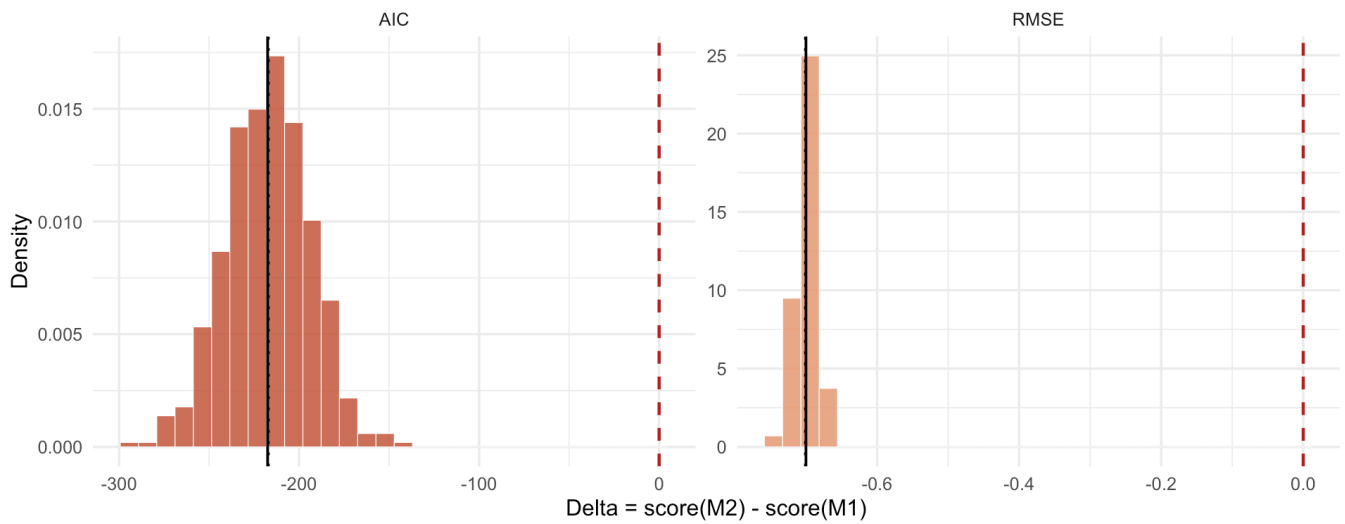
Prediction output	log_ratio	Jeffreys	CDF	Joint_Wald	Joint_DELR	AIC	RMSE
Without noise	1.000	1.000	1.000	1.000	1.000	-	-
With noise	0.668	0.668	0.706	0.668	0.668	-	-
Benchmark	-	-	-	-	-	1.000	1.000

Larger model true: DRM Delta distributions
Delta < 0 favours M2; Delta > 0 favours M1. Red dashed = 0; solid black = mean; dotted black = median.



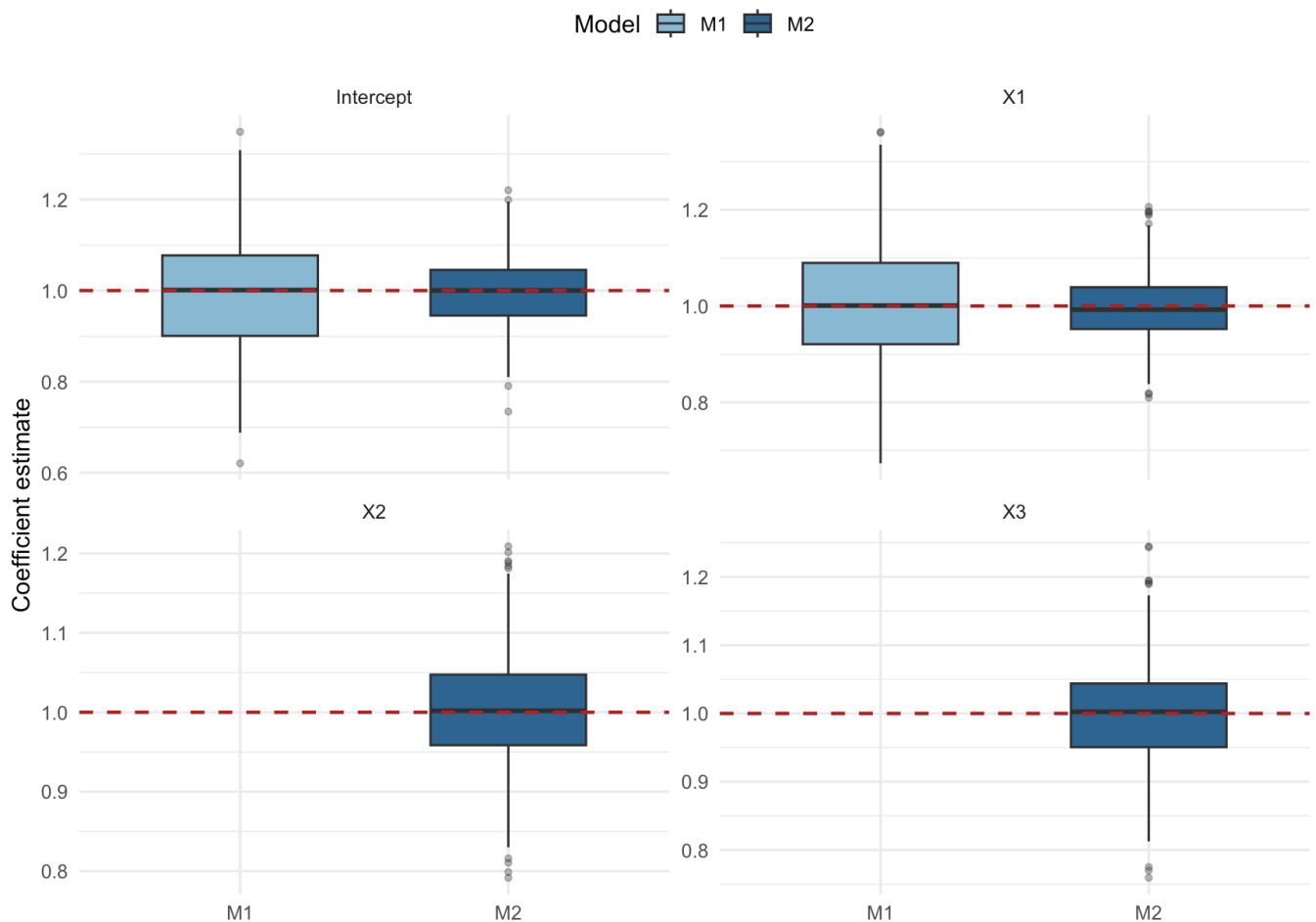
Larger model true: benchmark Delta distributions

Delta < 0 favours M2; Delta > 0 favours M1. Red dashed = 0; solid black = mean; dotted black = median.



Larger model true: coefficient estimates

Red dashed line = true coefficient



Part I: SNR experiments

Experiment 1A: the simpler model is true

$$Y = 1 + \beta_1 X_1 + \varepsilon,$$

Both candidate models contain the true signal. The shared SNR is

$$\text{SNR}_{\text{shared}} = \frac{\beta_1^2 \sigma_{x_1}^2}{\sigma_{\epsilon}^2}.$$

The shared SNR was also varied by changing the **predictor standard deviation** or the **observation-error standard deviation**. These three parameterisations produced essentially the **same selection pattern** at matched SNR values, so the **coefficient**-based implementation is used in the main results for clarity.

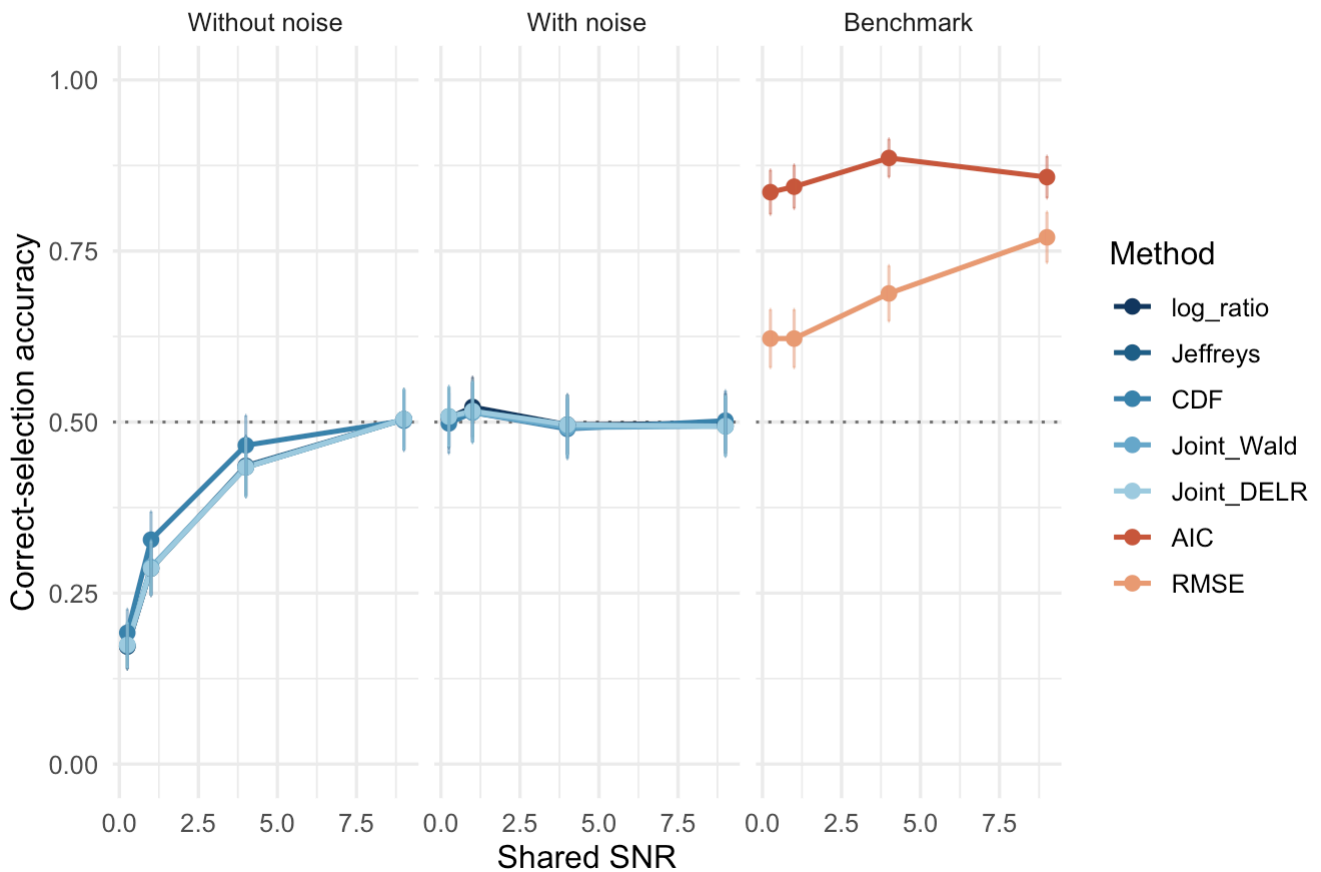
Three equivalent parameterisations of shared SNR

shared_snr implementation		beta_1	x1_sd	sigma_error
0.25	Coefficient	0.5	1.0	1.000
0.25	Predictor SD	1.0	0.5	1.000
0.25	Error SD	1.0	1.0	2.000
1.00	Coefficient	1.0	1.0	1.000
1.00	Predictor SD	1.0	1.0	1.000
1.00	Error SD	1.0	1.0	1.000
4.00	Coefficient	2.0	1.0	1.000
4.00	Predictor SD	1.0	2.0	1.000
4.00	Error SD	1.0	1.0	0.500
9.00	Coefficient	3.0	1.0	1.000
9.00	Predictor SD	1.0	3.0	1.000
9.00	Error SD	1.0	1.0	0.333

Conclu: Increasing this quantity strengthens signal shared by both candidates; it does not increase their structural difference. DRM methods have no explicit **complexity penalty**, so their accuracy may remain close to **chance level** even when shared SNR is high.

- **Simpler model true: accuracy by shared SNR**
- **Simpler model true: Delta by shared SNR**

Simpler model is true



Experiment 1B: the larger model is true

Here the larger model contains additional true signal. With unit predictor and error variances,

$$\delta = \sqrt{\text{SNR}_{\text{disc}}/2}.$$

The zero point is excluded because it belongs to Experiment 1A, where the simpler model is true.

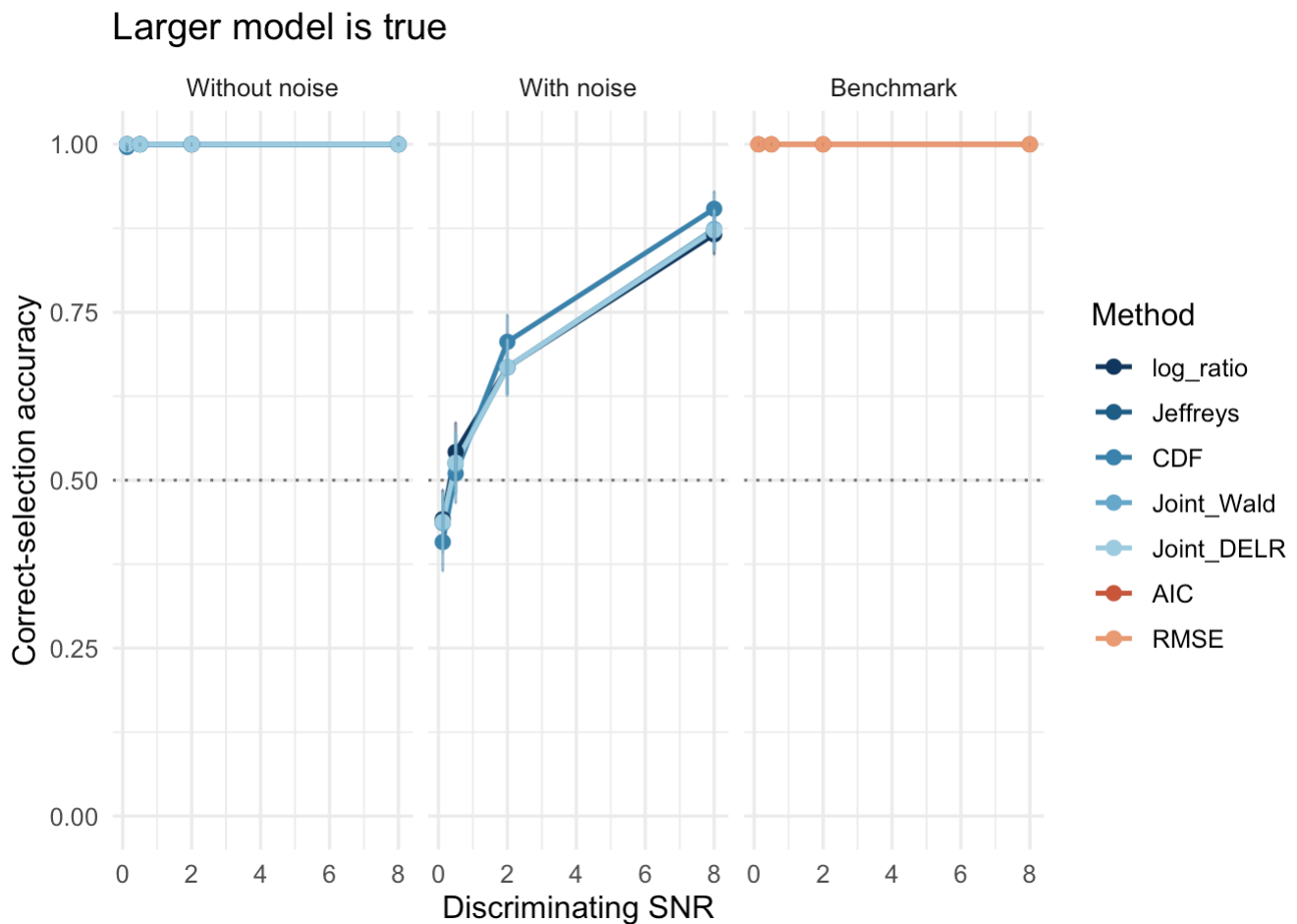
The same SNR equivalence applies here: changing the coefficient, predictor variance, or error variance produced essentially the same selection pattern at matched discriminating SNR values.

Conclu: When the larger model is true, with-noise DRM requires sufficient discriminating signal to identify the additional predictors, that is because after **predictive noise** is added, part of the signal omitted by the smaller model is absorbed into its estimated residual variance, allowing the misspecified model to produce a marginal predictive distribution that still resembles the observed response distribution.

Without-noise DRM and the two benchmarks show ceiling-level accuracy.

► **Larger model true: accuracy by discriminating SNR**

► **Larger model true: Delta by discriminating SNR**



Part II: distributional robustness

The robustness experiments are also separated by the true model. In both cases, all compared distributions have mean zero and variance one.

Concl: Overall, changing the **predictor or error distribution shape does not fundamentally alter the selection pattern.**

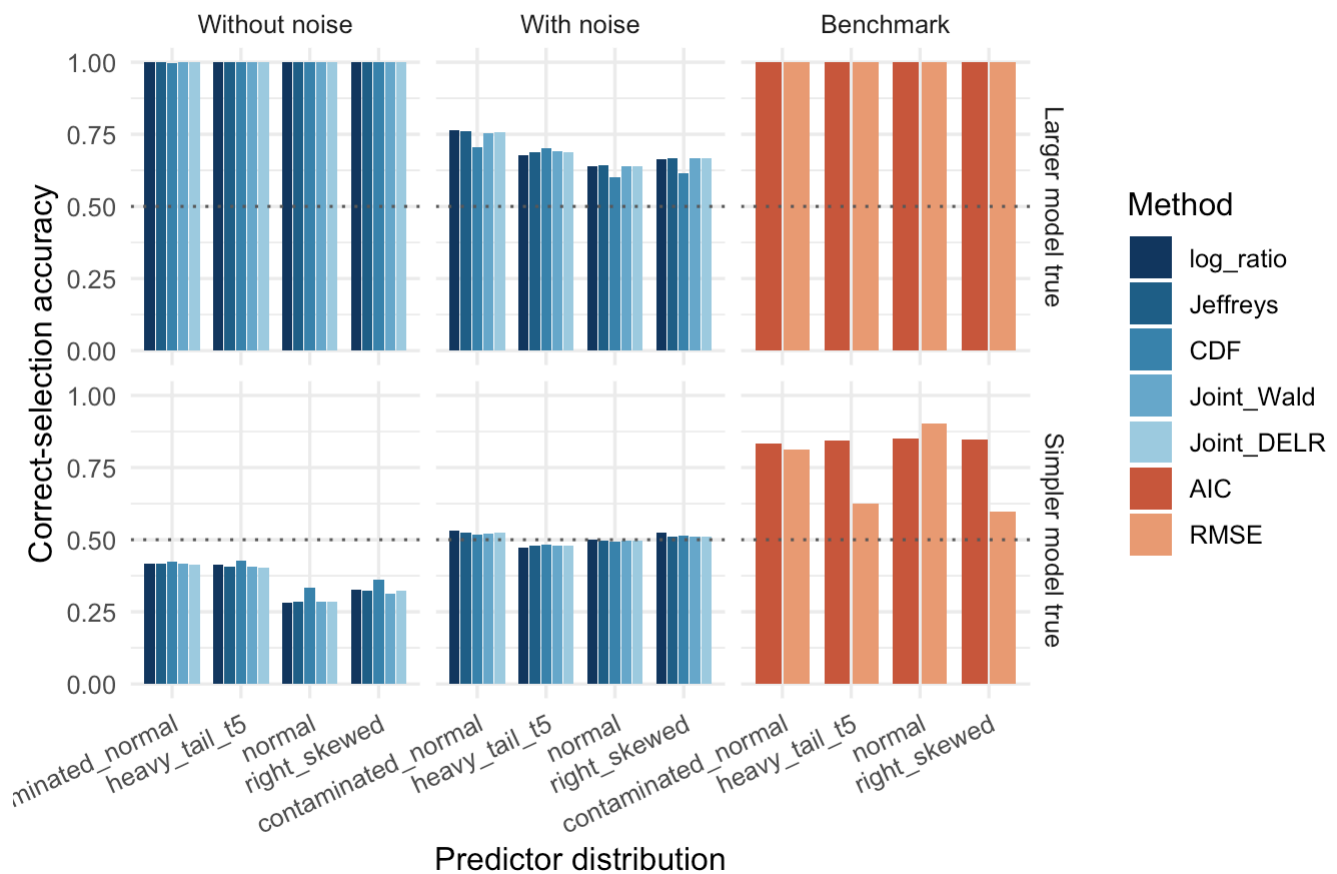
Experiment 2: changing the predictor distribution

For the simpler-true case, changing the distribution of the useless predictors examines finite-sample overfitting. For the larger-true case, the discriminating SNR is fixed at

$$\text{SNR}_{\text{disc}} = 2.$$

- Accuracy under different predictor distributions
- Delta under different predictor distributions

Predictor-distribution robustness



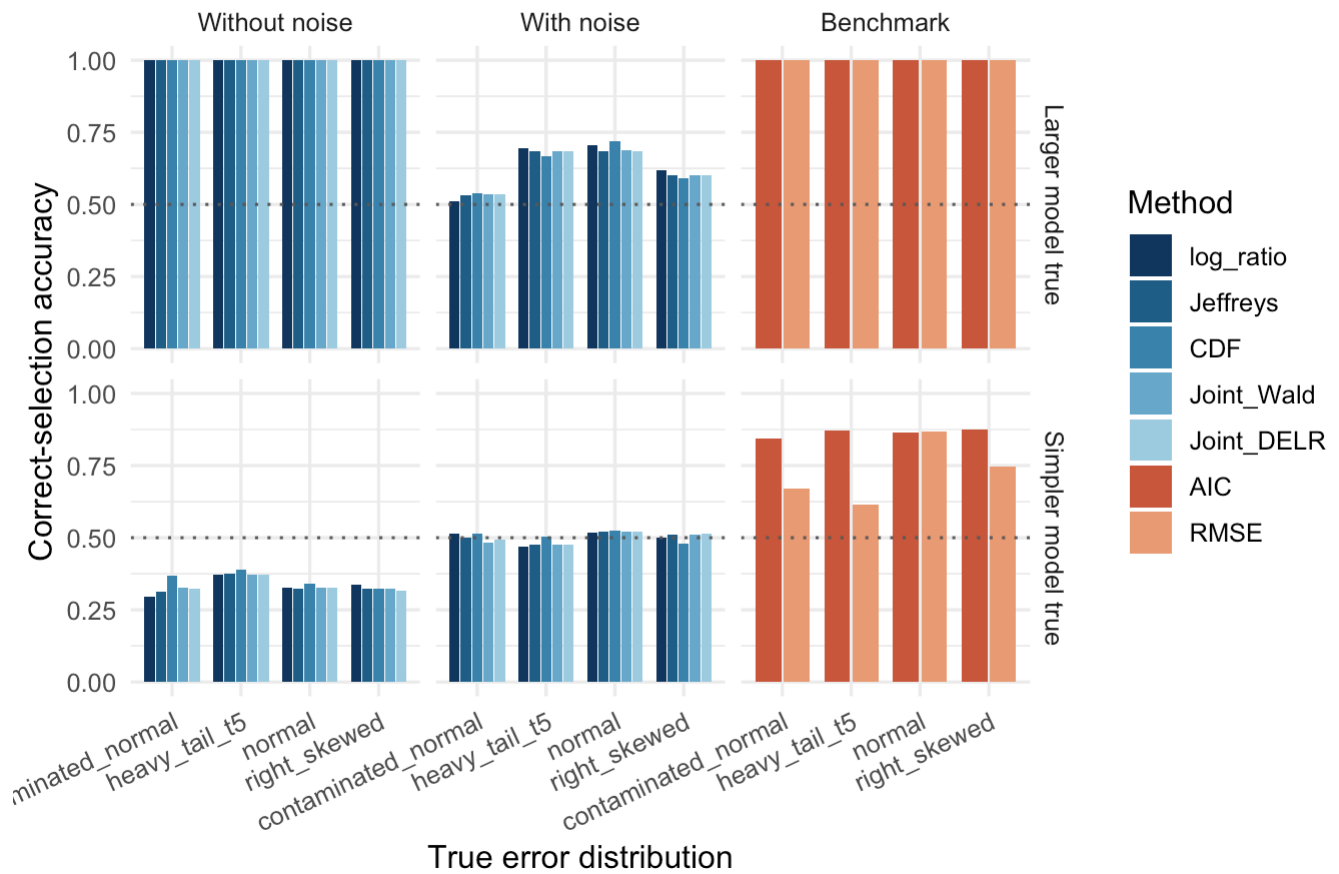
Experiment 3: changing the error distribution

Every true error distribution has mean zero and variance one. Model-generated predictive errors (noise) remain standard-normal in every scenario. Where the fitted linear model does not know the true error family.

► Accuracy under different error distributions

► Delta under different error distributions

Error-distribution robustness



Non-nested model

We compare two non-nested candidate models:

$$M_1 : Y \sim X_1 \quad vs \quad M_2 : Y \sim X_2 + X_3.$$

$$X_1, X_2, X_3 \stackrel{\text{iid}}{\sim} N(0, 1), \quad \varepsilon \sim N(0, 1).$$

$$M_1 : Y = 1 + \beta_1 X_1 + \varepsilon, \quad \text{SNR} = \beta_1^2.$$

$$M_2 : Y = 1 + \delta X_2 + \delta X_3 + \varepsilon, \quad \text{SNR} = 2\delta^2.$$

► Non-nested accuracy by SNR

